ASSIGNMENT 3: CAUSALITY ASSESSMENT

1. Introduction

Causality assessment is the process of determining whether a suspected adverse drug reaction (ADR) is related to a particular drug. It helps evaluate the likelihood of a causal relationship between drug administration and an adverse event.

Two commonly used methods are the *Naranjo Scale* and the *WHO-UMC* causality assessment system.

2. Case Summary

Case title: Metformin-Induced Generalized Bullous Fixed-Drug Eruption

A 43-year-old male developed recurrent generalized bullous skin lesions after taking metformin for weight loss.

- Suspected drug: Metformin
- Indication: Weight loss
- Duration of exposure: Approximately 5 years, with irregular use
- Onset: Skin lesions first appeared around two weeks after starting metformin.
- Clinical diagnosis: Generalized bullous fixed-drug eruption
- Investigation: Skin biopsy findings supported the diagnosis.
- Dechallenge: The progression of lesions stopped after metformin was discontinued.
- Rechallenge: The lesions reappeared more severely after the patient restarted metformin.

3. Factors Supporting Causality

- Drug-related: Previous reports have described metformin-associated fixed-drug eruptions.
- Event-related: The patient developed generalized bullous skin lesions.
- Patient-related: No underlying disease or other concomitant medicines were reported.
- Clinical evidence: The reaction improved after drug withdrawal and returned after re-exposure. Biopsy findings supported the diagnosis.

4. **Naranjo ADR Probability Scale**

The Naranjo Algorithm consists of 10 questions used to assess the probability that an adverse reaction is related to a suspected drug.

Sl.No.	Question	Yes	No	Dont Know	Score
1	Are there previous conclusive reports on this reaction?	+1	0	0	+1
2	Did the adverse event appear after the suspected drug was administered?	+2	-1	0	+2
3	Did the adverse reaction improve when the drug was discontinued or a specific antagonist was administered?	+1	0	0	+1
4	Did the adverse reaction reappear when the drug was readministered?	+2	-1	0	+2
5	Are there alternative causes that could independently have caused the reaction?	-1	+2	0	+2
6	Did the reaction reappear when a placebo was given?	-1	+1	0	0
7	Was the drug detected in the blood or other fluids at toxic concentrations?	+1	0	0	0
8	Was the reaction more severe when the dose was increased or	+1	0	0	0

	less severe when the dose was decreased?				
9	Did the patient have a similar reaction to the same or similar drugs in any previous exposure?	+1	0	0	+1
10	Was the adverse event confirmed by any objective evidence?	+1	0	0	+1

TOTAL SCORE: 10

Interpretation of Naranjo Score:

- 9 or above – Definite
- 5 to 8 – Probable
- 1 to 4 – Possible
- 0 or below – Doubtful

Result: The total Naranjo score is 10, indicating a **Definite** ADR.

5. **WHO-UMC Causality Assessment**

The WHO-UMC system classifies the relationship between a suspected drug and an adverse reaction into six categories:

Category	Description
Certain	A clear temporal relationship, plausible reaction, and convincing evidence such as positive dechallenge and rechallenge.
Probable/ Likely	A reasonable time relationship, unlikely alternative causes, and clinically reasonable response to drug withdrawal.
Possible	A reasonable time relationship, but other causes may also explain the reaction.

Unlikely	A time relationship makes a causal connection improbable.
Conditional/ Unclassified	More information is required, or additional data are being examined.
Unassessable/ Unclassifiable	Available information is insufficient or contradictory and cannot be supplemented.

Assessment of the case: **Certain**

The case demonstrates a clear temporal relationship, improvement after discontinuation of metformin, and recurrence of the reaction after re-exposure. These findings strongly support a causal relationship.

6. Conclusion

Causality assessment helps establish the relationship between a suspected drug and an adverse drug reaction.

In this case, the Naranjo Scale produced a score of 10, indicating a definite adverse drug reaction. The positive dechallenge and rechallenge findings also strongly support the association between metformin and generalized bullous fixed-drug eruption.

7. References

Abtahi-Naeini B, et al. Metformin-Induced Generalized Bullous Fixed-Drug Eruption with a Positive Dechallenge-Rechallenge Test: A Case Report and Literature Review. Case Reports in Dermatological Medicine. 2023